

The Powder Tax: The Real Bottleneck in Utah's Ski Traffic Isn't on the Road

By Adam Looney

Before I learned he was a Snowbird legend and I adopted him as my barometer for how long the lines were getting, he was just a guy we called Dapper Dan. Everyone else in the parking lot is wearing technical fabrics—three-layer Gore-Tex, merino wool blends, heated gloves—and Dan is booting up in a button-down shirt. The collar sticks out from under his jacket, and when his shirt tails get untucked, they peek out when he sits on the lift. He leaves a white straw hat with a black band on the dash of his Highlander, after swapping it for his helmet.

Dan is a legend because season after season, he racks up millions of vertical feet at the 'Bird. He lines up at 8:15. Most days, that makes him first in line. And then he skis bell-to-bell. If you happen to ride up with him—and, odds are, you eventually will—you can ask him how he does it. His formula is simple. There are no breaks. He eats a sandwich from his pocket while riding the chairlift. If it's crowded, he takes the singles line. Dan doesn't buy "Fast Tracks" or have a special pass that lets him cut the line. He is in the trenches with the rest of us.

We know he skis millions of vertical feet because every day, he logs into the Snowbird app which records every lap and vertical foot of skiing. He's always in the top 25 of the leaderboard, which means his statistics are a public record each day.

That makes him to daily skiing conditions what a "standard candle" is to astrophysics—a star of known luminosity used as a benchmark when measuring the cosmos.

As a skier trying to understand the modern crowding crisis, Dan is my standard candle for congestion.

And lately, what he's been measuring is the same thing every skier in the Wasatch has been living.

Utah — like many ski destinations — is in the midst of a congestion crisis. In Little Cottonwood and Big Cottonwood, the misery starts long before your first turn: stopped traffic on powder mornings, neighborhoods clogged by idling cars, the "red snake" of tail lights inching toward the resorts and back down again. In response, the state has begun treating skiing like an infrastructure problem — debating wider roads, dedicated bus lanes, and even a massive gondola projected to cost upwards of \$700 million — all to get more people into the canyons.

But Dan's numbers suggest we've misdiagnosed the bottleneck. The worst congestion isn't on the road. It's on the mountain—paid not in dollars, but in vertical feet lost to lift lines. On weekends, when the lot is full, and especially on powder days, the number of laps Dan gets plummets.

Because Dan's effort is constant — he never stops, never tires, and never lingers in the lodge — these daily variations in his vertical are not a measure of his effort or athleticism. They are a measure of the mountain's inefficiency. Every vertical foot Dan doesn't ski is a foot he spent waiting in line.

Last season, every day from January to April, I opened the app and wrote down exactly how many vertical feet he skied that day. I tracked Dan, the rest of the Snowbird leaderboard, plus leaderboards at Alta and Solitude for 90 days, 3,600 observations, and roughly 2,100 unique skiers.¹

The resorts that impose real marginal costs or hard limits—like Alta's weekend parking restrictions and Solitude's daily parking charges—don't eliminate the crush, but they measurably blunt it.

The Standard Candle, Quantified

On a typical weekday, Dan skis 57,500 vertical feet. His best days — usually a Tuesday or Wednesday — top 59,000. His peak for the season: 67,870 feet. At Snowbird, with roughly 3,000 vertical feet per tram lap, that's like twenty top-to-bottom runs.

But his output isn't constant. And the variation is the point: it's a measure of congestion, not effort.

The Weekend Tax: Dan averages 59,600 feet on a Tuesday. On a Saturday, that drops to 49,700—nearly 10,000 feet, or about three tram laps fewer. Same skier, same routine. More waiting. On average, on Friday, Saturday, and Sunday he gets 10% fewer laps than during the work week.

The Lot-Full Tax: The clearest marker is when Snowbird's parking lot hits capacity and they tweet out "Parking at Snowbird is FULL." On those days, Dan's vertical falls from 57,500 to 45,200—down more than 12,000 feet, about 21% fewer runs.

The Powder Penalty: The most distressing pattern shows up when conditions are best. Utah sells itself on storms — "The Greatest Snow on Earth," is stamped on the license plates — but the data shows that snowfall draws a crowd large enough to overwhelm the benefit of better snow.

On a clear day with no snowfall, Dan averages nearly 60,000 vertical feet. After a solid storm — an inch and a half of water equivalent, roughly 15 to 20 inches of fresh snow — he averages 47,200. But on the deepest cycles of the season, when the Wasatch picks up 20

¹ I tracked the Snowbird, Alta, and Solitude leaderboards daily from January 18 through April 20, 2025. I skipped Brighton because I worried night skiing distorted the numbers. I only forgot to check three days, which I think was pretty good. The Snowbird app tracks skiers at Snowbird; the Ikon app tracks skiers across all three resorts. In total, the dataset includes 3,600 daily skier observations—the top 10 ranked skiers at each resort each day.

to 30 inches or more over multiple days—the days when all of Salt Lake seems to call in sick — his output collapses to roughly 32,000. On what should be the best days, *he spends half his time waiting in line*.

It's Not Just Dan

A natural question: Is Dan an anomaly? Maybe the standard candle is flickering. Maybe he just happens to ski less on crowded days for reasons unrelated to congestion.

He isn't. I recorded the top ten skiers on the Snowbird app leaderboard over the course of the season. These are not casual skiers. They are the most hardcore skiers and boarders at the resort on any given day, and are pretty protective of their leaderboard status. When I run the same analysis across all of them, observing their individual variability from day to day, the patterns hold.

On “Lot FULL” days, the average leaderboard skier's vertical drops from 43,500 to 36,000 (17%). The weekend penalty is about 10%. Every additional foot of snow means people wait in line about 18% longer.

What's happening isn't subtle. It isn't about one skier's bad luck or personal choices. The mountain itself is running out of throughput. On a day when the lot fills up, every skier on the leaderboard — regardless of fitness, skill, or strategy — skis less. The congestion is systemic.

A Tale of Three Resorts

Dan's data is from the Snowbird app, which only tracks Snowbird. But the Ikon pass app tracks skiers across multiple resorts, including Snowbird, Alta and Solitude. This lets us answer a crucial question: Is this a Snowbird problem, a Little Cottonwood problem, or a Wasatch-wide problem?

It turns out it's a mix—and the differences between resorts are revealing.

The weekend penalty is largest at Snowbird, small at Alta (about 5%), and at Solitude, leaderboard skiers there ski 6% *more* on weekends. Why? Alta requires reservations on Friday, Saturday, and Sunday, and Solitude imposes paid parking all week. The data is suggestive that the parking rules are effective in smoothing out demand.

Likewise, the Lot Full effect is universal but particularly severe at Snowbird. When Snowbird's parking lot fills up—a proxy for a big day across the canyons—vertical drops at every resort. Alta: 10%. Even Solitude, over in Big Cottonwood Canyon, sees a 4% drop. The crush of a canyon-wide powder day reaches every lift line in the Wasatch.

Most importantly, while each foot of snow falling on Snowbird crushes the number of laps by about 20%, at Solitude the congestion penalty is only 9%. And Alta? Skiers there get in the same amount of vertical regardless of how much it snows.

This is the most telling finding in the data. Snowbird and Alta share a canyon, share a road, share the same storms. The difference is management. Alta has fewer parking spots; on weekends, it requires reservations, and it doesn't accept skiers with the Ikon Base Pass. Likewise, Solitude has daily paid parking. These systems are effective at reducing congestion. While these resorts do get crowded on weekends and busy days, their attempts to deal with congestion with parking rules seem like they're working.

Snowbird, by contrast, lets the lot fill until it's bursting, then posts a sign on the highway.

The Economics of Congestion

This suggests that congestion is not an infrastructure problem; it's a pricing — or rationing — problem, and one directly caused by skiing's new megapass-driven business model. Today most skiers have a season pass rather than buying day tickets. When everyone already has a season pass, an extra ski day is free. The system still has to ration access somehow. In Utah, and at many resorts across the U.S., that rationing happens in time: lift lines, gridlock, and the slow crawl of the Red Snake.

It is under appreciated how much resort pricing has changed. For decades, season passes were luxury goods: a small number of committed locals paid a steep price for unlimited access, while everyone else bought relatively cheap day tickets. Then a wave of consolidation flipped the model. Two companies — Vail Resorts (Epic Pass) and Alterra Mountain Company (Ikon Pass) — acquired or affiliated with resorts across the country until they controlled more than half of all U.S. lift capacity. In Utah, seven of fifteen resorts — including most of the largest — are now on one network or the other, covering roughly three-quarters of the state's lift capacity.

The duopoly's competitive weapon wasn't higher prices; it was lower ones. When Vail acquired Stowe for the 2017–2018 season, the price of a season pass dropped from \$2,313 to \$899 — and the new pass was good at every other Epic resort in the country. Similar shocks rippled through Solitude, Sugarbush, Killington, and dozens of other mountains as they joined mega-pass networks. Even independent resorts had to cut season-pass prices to compete. Adjusted for inflation, season-pass prices across New England have fallen steadily since the mid-2010s. Day tickets, meanwhile, moved in the opposite direction: up 60% in real terms since the 1990s, with the steepest increases in the last decade.

The result is a system that's great for anyone who skis enough days to justify a pass — and that's a lot of people. Pass sales have surged. But that change had a brutal side effect. Once you've paid for a season pass, every additional day on the mountain is free. You don't weigh the cost of a lift ticket against the weather forecast. You just go. That's not a moral failing; it's the most basic prediction in economics. When the price of something falls to zero, people consume more of it. Total visits to ski resorts have climbed, driven increasingly by a core of season-pass holders skiing more and more days — because every additional day is free.

That's the origin of the congestion crisis. The congestion is the new day ticket — it's just collected in time and misery rather than dollars. Which makes the most perverse twist all the more galling. Having created the conditions for unbearable lines, the resorts now sell you a way out. Snowbird's "Fast Tracks" program charges \$89 per day — or \$6,750 for the season-long "Base Camp" upgrade — to skip the line entirely. The worse the experience gets for the average skier, the more valuable the premium product becomes. The resort profits not by fixing congestion, but by selling an escape from it. Taxpayers shouldn't be spending hundreds of millions on a gondola to deliver more customers to a system built on this incentive structure.

The Policy Trap

This brings us back to the billion-dollar question: What do we do about it?

The current policy debate is fixated on “uphill capacity”—how to get more people up Little Cottonwood Canyon faster. Proponents of the gondola argue that it will solve the traffic crisis.

But Dan's data suggests this gets the problem exactly backwards. If you uncap the road without expanding the mountain, you are merely moving the bottleneck from the asphalt to the snow. Even if UDOT built the gondola tomorrow, it would only worsen the congestion at the top. The lifts are already full. Each new carload or gondola cabin that arrives at the base subtracts a few thousand vertical feet from every other skier's day. You'll just have more people stuck in longer lift lines instead of in their cars.

The uncomfortable truth is that we need fewer people in the canyons at peak times, not more. And the data shows this clearly. The comparison between Snowbird, Solitude, and Alta amounts to a natural experiment: two resorts, same canyon, same snow, same road—one limits access, one doesn't. The resorts that limit access show no weekend penalty and smaller congestion costs.

The good news is that the solutions are straightforward, and some are already working. Alta and Solitude prove that parking limits and reservations dramatically reduce congestion. The data is unambiguous: their skiers experience almost none of the weekend penalty that defines a season at Snowbird. Snowbird should follow their lead.

Congestion pricing could go further. Utah already runs demand-based tolling on I-15, and Millcreek Canyon charges a modest fee that reduces congestion and funds maintenance — this isn't hypothetical. Apply the same principle to Little Cottonwood and Big Cottonwood: toll the canyons on powder days and peak weekends, exclude public transit, discount carpools and advance purchases, and use the revenue to fund expanded bus service, trail maintenance, and off-peak discounts that make skiing more accessible, not less. We dread the idea of paying more to drive up the canyon, but the reality is we're already paying — in parking fees, higher ticket prices, and Fast Tracks programs designed to monetize the misery. The only question is whether those costs flow to resort owners or back to the public.

And some resorts should be bold enough to leave the mega-passes altogether. Reestablish a season pass that covers costs without relying on megapass membership, and let the day-to-day price of skiing do what prices are supposed to do: signal scarcity. Twenty years ago, this was how the industry worked. It can work again.

Beyond demand, there's supply. There has been almost no growth in the number of U.S. ski resorts in twenty-five years, and relatively little expansion of terrain at existing ones. More terrain — whether new resorts, expanded boundaries, or additional lift capacity — is the only way to grow the pie rather than just ration it.

These ideas require accepting a truth that the canyon's boosters would rather not face: the Wasatch has a physical limit, and we've already blown past it. But accepting that limit is what makes the solutions possible. A gondola that delivers more people to the same chokepoints isn't a fix. Pricing that spreads demand, limits that protect the experience, and terrain that expands capacity — those are.

Dan will keep skiing bell-to-bell either way. The rest of us would just like a few more of those vertical feet back.